

PRODUCT REQUIREMENTS DOCUMENT

CineChain

Decentralized Film Distribution Protocol

Open Source • Blockchain-Powered •
Creator-First

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Document Information

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0.1	Initial draft — problem framing and vision
0.5	Added watermarking architecture and smart contract model
0.8	Incorporated governance DAO structure and credits economy
1.0	Full PRD — tech stack, architecture diagrams, roadmap, risks

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1. Executive Summary

CineChain is an open-source, blockchain-based film distribution protocol that eliminates the structural dependency of filmmakers and independent distributors on centralized streaming intermediaries. The platform enables any rights holder to upload, tokenize, and distribute a film directly to a global audience — with real-time, auditable revenue distribution via smart contracts, session-level forensic watermarking to deter and attribute piracy, and a credits-based economic layer that abstracts cryptocurrency complexity from mainstream audiences.

Every film on CineChain is registered on-chain as a sovereign content asset. Viewing rights are issued as Film License Tokens (FLTs) in three tiers — Rental, Ownership, and Collector — allowing audiences to engage with content in ways that match their intent and willingness to pay. Revenue flows automatically to every rights holder defined in the smart contract at the moment of upload, with no intermediary required and no audit delay.

The platform is governed as a decentralized autonomous organization (DAO), ensuring that no single entity — including the founding team — can unilaterally alter fee structures, delist content, or restrict access to audience data. CineChain's value proposition to the filmmaking community is structurally enforced by code, not by corporate policy.

CineChain's mission: to make film distribution as open, transparent, and creator-aligned as the creative process itself.

Key Metric	Current Industry State	CineChain Target
Revenue transparency	12–18 month reporting delay	Real-time on-chain
Distributor take rate	30–50% of gross revenue	5% protocol fee
Piracy attribution	Rarely possible	Per-session forensic ID
Filmmaker data access	Controlled by platform	Fully owned by creator
Governance model	Corporate, opaque	DAO, public voting

2. Problem Statement

2.1 The Intermediary Capture Problem

The global film distribution industry is structurally dominated by a small number of powerful intermediaries — major streaming platforms, theatrical exhibitors, and traditional distributors — who collectively control the pricing, algorithmic visibility, audience data, and commercial fate of virtually every film released. For the independent filmmaker or mid-tier distributor, this creates a dependency relationship that is economically asymmetric and operationally opaque.

Netflix, Amazon Prime Video, Disney+, and Apple TV+ collectively account for the overwhelming majority of paid streaming revenue globally. Access to their platforms requires negotiation, exclusivity agreements, and acceptance of terms that are not publicly disclosed and are essentially non-negotiable for any counterparty without significant market leverage. The platform retains control of pricing, curation placement, marketing spend, territory rights, and — most critically — the raw performance data that would allow a filmmaker to independently assess how their work is actually performing commercially.

2.2 Revenue Opacity and Reporting Fraud

Revenue reporting in the film industry is notoriously unreliable. Traditional distribution agreements require the distributor to produce quarterly or annual revenue statements, but the methodology for calculating net receipts — after deducting distribution expenses, marketing costs, overhead allocations, and other line items — is both complex and largely unauditable by the rights holder. Multiple high-profile lawsuits from major studio talent in recent decades have established that even films generating hundreds of millions of dollars in gross revenue can be reported as having no net profits under standard Hollywood accounting practices.

Independent filmmakers face the same structural problem at a smaller scale: they receive periodic statements they cannot verify, from distributors they cannot audit, on timelines that are contractually defined but rarely enforced. The practical consequence is that a significant portion of the revenue generated by independent film never reaches its creators.

2.3 Margin Compression

Beyond opacity, the absolute level of revenue extracted by intermediaries is structurally harmful to the independent film ecosystem. Streaming platforms typically pay between 20% and 70% of a defined licensing fee, with the specific amount determined by negotiations that heavily favor the platform. Theatrical distributors retain between 30% and 50% of box office gross before passing receipts to the production. When layered with sub-distribution agreements, sales agent commissions, and producer fees, the actual revenue reaching the filmmaker can be a fraction of what the content generated commercially.

2.4 The Piracy Attribution Gap

Piracy costs the global film industry an estimated \$29 billion annually in lost revenue, with the majority of that loss concentrated in the period immediately surrounding a film's initial release window — the window that most directly determines its commercial trajectory. Current anti-piracy approaches fall into three broad categories: DRM-based access control (which is

routinely defeated by technical users), watermarking (which exists in enterprise form but is prohibitively expensive for independent distributors), and takedown enforcement (which operates reactively, after millions of unauthorized views have already occurred).

The fundamental gap in the current landscape is forensic attribution at the source. When a pirated copy of a film surfaces online, identifying which screener, which platform session, or which theater screening produced the unauthorized copy is technically possible in theory but operationally difficult in practice. The result is that the enforcement mechanism is blunt, slow, and rarely deters determined actors.

2.5 Content Discovery Failure

The consolidation of distribution into a small number of algorithmically-governed platforms has created a discovery problem that is as damaging to independent film as any of the economic issues described above. Algorithmic curation on major platforms is optimized for engagement metrics — watch time, completion rate, and subscription retention — that structurally favor high-budget, broadly accessible content. Independent and arthouse films, international productions, and debut features are systematically underrepresented in recommendation surfaces despite often serving audiences with high engagement and loyalty.

The practical consequence is that a significant portion of the global film catalog is commercially invisible, not because it lacks an audience, but because the infrastructure designed to connect content to audiences is misaligned with its needs.

Problem Summary

1. Revenue opacity — filmmakers cannot verify or audit what they are owed
2. Margin compression — intermediaries extract 30–50% of gross revenue
3. Piracy attribution gap — unauthorized redistribution is rarely traceable to source
4. Content discovery failure — algorithmic curation buries independent film
5. Data ownership deficit — audience data is held exclusively by platforms
6. Governance asymmetry — platform rules are opaque and unilaterally changeable

3. Vision & Mission

3.1 Vision Statement

A world where every filmmaker can reach every audience directly — with full transparency, full revenue accountability, and complete creative sovereignty.

3.2 Mission Statement

CineChain builds the open-source protocol infrastructure that enables film to be distributed, monetized, and experienced as a sovereign creative asset — independent of any intermediary, governed by its community, and protected against unauthorized reproduction at every layer of the delivery stack.

3.3 Design Principles

- Creator sovereignty: the rights holder retains complete control over pricing, access terms, territory, and revenue splits from the moment of upload.
- Radical transparency: every transaction, revenue distribution, and governance vote is publicly verifiable on-chain.
- Structural neutrality: the protocol cannot favor any distributor, studio, or platform over another. Curation is community-driven, not algorithmic.
- Mainstream accessibility: blockchain infrastructure must be invisible to the casual viewer. Complexity is absorbed by the protocol layer, not imposed on the audience.
- Forensic accountability: every copy of every film is uniquely attributed to its authorized recipient. Unauthorized redistribution is technically attributable and legally actionable.
- Open-source permanence: the protocol is publicly readable, forkable, and improvable. No entity can unilaterally close or capture it.

4. Target Audience & User Personas

4.1 Primary User Segments

Segment 1: Independent Filmmakers and Distributors

This segment encompasses directors, producers, and small independent distributors who have completed films and are seeking viable distribution routes outside of the major platform ecosystem. They typically have limited negotiating leverage with streaming platforms, limited budgets for marketing and distribution infrastructure, and a strong interest in maintaining creative and commercial control over their work.

Their primary pain points are revenue opacity, slow payment cycles, and the inability to independently reach their audience without a platform intermediary. CineChain directly addresses all three.

Segment 2: Cinephile Audiences and Film Enthusiasts

This segment consists of viewers with above-average engagement with film as an art form — festival attendees, subscribers to niche streaming platforms like MUBI or Fandor, active participants in film communities, and collectors of physical media. They are comfortable paying a premium for access to content they value, are interested in ownership and provenance, and are often early adopters of new consumption models.

They represent the demand side of the platform and are the most natural early adopters of the Film License Token model, particularly the Ownership and Collector tiers.

Segment 3: Film Festivals and Curatorial Organizations

Film festivals, particularly those operating at the independent and international tier — Sundance, SXSW, Tribeca, IDFA, and their equivalents globally — represent a critical institutional partner segment. They provide pre-curated content catalogs, established audience relationships, and curatorial credibility that can accelerate both supply-side content acquisition and demand-side discovery.

Segment 4: Film Schools and Educational Institutions

Educational institutions represent a distinct institutional buyer segment with specific access requirements — multi-user licensing, course integration, archival access, and provenance verification. CineChain's smart contract infrastructure can accommodate bulk access licenses issued directly to institutions, with revenue flowing automatically to rights holders.

4.2 User Personas

Persona	Profile & Needs
Aisha — Independent Filmmaker	Age 34, Lagos. Has completed her second feature. Sold North American rights to a mid-tier distributor who has paid her once in three years. Needs: transparent revenue tracking, direct audience access, simple upload workflow.

Marco — Cinephile Collector	Age 41, Milan. MUBI subscriber, owns 200+ Blu-rays. Wants to own and resell films digitally with verifiable provenance. Needs: collector-tier tokens, resale market, curation discovery.
SunFilm Festival — Curatorial Partner	Regional arthouse festival. Wants to extend digital reach post-screening. Needs: curatorial badge system, revenue share on digital sales, audience data for curated titles.
Dev — Crypto-Native Viewer	Age 27, Singapore. Comfortable with wallets, DeFi, and NFTs. Wants to invest early in films that may appreciate. Needs: collector tokens, secondary market, on-chain provenance.
Priya — Casual Viewer	Age 31, Mumbai. Watches 2–4 films monthly. No crypto experience. Needs: credit card purchase, simple UX, no wallet complexity, CinePass subscription option.

5. Solution Overview

CineChain is structured as three interdependent layers: a blockchain protocol layer that handles content registration, token issuance, and smart contract execution; an application layer that provides the filmmaker upload workflow, audience-facing storefront, and credit economy; and a governance layer that manages protocol upgrades and community curation. These three layers are designed to be independently operable and composable — third-party developers can build on the protocol layer without using the CineChain application, and the governance layer can upgrade the protocol without disrupting existing content or transactions.

Layer	Function	Key Components
Protocol Layer	On-chain infrastructure	Content hash registry, FLT smart contracts, revenue distribution contracts, DAO governance contracts
Application Layer	User-facing products	Filmmaker dashboard, audience storefront, credit economy, streaming delivery, watermarking engine
Governance Layer	Protocol management	DAO voting, curator badge system, protocol upgrade proposals, fee adjustment voting

6. Core Features — Detailed Specification

6.1 Film Upload and On-Chain Registration

When a rights holder uploads a film to CineChain, the upload workflow proceeds through four sequential stages: content validation, encryption, hashing, and on-chain registration.

Content Validation

The filmmaker uploads the film file along with a structured metadata package that includes title, director, year, runtime, genre classifications, territory rights, content rating, and credit information. The platform validates the file format (accepting DCP, ProRes, H.264, H.265, and AV1), checks for minimum quality thresholds, and cross-references the title against the on-chain content registry to prevent duplicate registrations.

Encryption

Upon validation, the film is encrypted using AES-256-GCM with a unique content encryption key (CEK) generated for each film. The CEK itself is encrypted using the rights holder's public key and stored in a key management service. At the point of authorized viewing, the CEK is decrypted and delivered to the viewer's authorized session — never transmitted in cleartext. The encryption envelope is constructed such that stripping it without the authorized key produces an unviewable output.

Hashing and On-Chain Registration

A SHA-3-256 hash of the unencrypted master file is computed before encryption and registered on the blockchain alongside the film's metadata package. This hash serves as a permanent, immutable proof of the film's original state. Any subsequently surfaced copy — regardless of re-encoding, compression, or alteration — can be compared against this reference hash to determine whether it originated from the CineChain platform and whether it has been modified. The registration transaction is signed by the rights holder's wallet and emits an event that permanently anchors the film's provenance on-chain.

6.2 Film License Token (FLT) Model

Film License Tokens are ERC-1155 compatible smart contracts that represent specific, legally defined viewing rights. The choice of ERC-1155 over ERC-721 enables both fungible (multiple identical rental tokens) and non-fungible (unique collector editions) token types within a single contract, reducing gas costs and simplifying the smart contract architecture.

Token Tier	Rights Granted	Transferability
Rental Token	48-hour streaming access from first play. No download. Session-watermarked delivery.	Non-transferable. Expires after use window.
Ownership Token	Perpetual personal streaming access. No download. Session-watermarked on each view.	Transferable once to another wallet (gift). Not resalable on secondary market.

Collector Token	Perpetual access + verifiable provenance record + secondary market resale rights. Eligible for filmmaker-signed editions.	Fully transferable and resalable. Resale triggers revenue share to original rights holder.
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Secondary Market and Resale Royalties

Collector Tokens can be listed for resale on CineChain's integrated secondary market. Every resale transaction triggers the original smart contract, which automatically distributes a resale royalty — typically 10% of the resale price — to the rights holder defined at the time of minting. This royalty is encoded in the contract and cannot be bypassed even if the token is transferred to an external marketplace that supports ERC-1155 royalty standards (EIP-2981). For the first time, filmmakers receive an ongoing economic stake in the secondary appreciation of their work.

6.3 Credits Economy

All consumer-facing transactions on CineChain are conducted in CineCredits — a platform-level stable unit of account pegged to USD at a fixed rate of 1 CineCredit = \$0.10. CineCredits are purchased through standard payment rails including Visa, Mastercard, PayPal, and Apple/Google Pay, as well as directly via cryptocurrency for users who prefer it. The credits layer serves three critical functions in the platform architecture.

- **Price stability:** content pricing in CineCredits is unaffected by cryptocurrency market volatility, making the platform commercially predictable for both filmmakers and audiences.
- **UX abstraction:** casual users never interact with wallets, gas fees, or token mechanics. They purchase credits with a card and spend them on films, exactly as they would purchase tokens on a gaming platform or App Store.
- **On-chain auditability:** despite the consumer-facing simplicity, every credit redemption is recorded on-chain as a token issuance event, preserving the full transparency and auditability of the underlying blockchain infrastructure.

CinePass — The Subscription Tier

CineChain offers a monthly subscription tier called CinePass for viewers who prefer the predictability of a fixed monthly payment. CinePass subscribers pay a fixed monthly fee and receive a pre-allocated bundle of CineCredits sufficient for approximately four to six standard rental transactions per month. Unused credits roll over for one month. CinePass operates on the same underlying token infrastructure as individual purchases — each film viewed by a CinePass subscriber still generates a rental token issuance event and triggers the same smart contract revenue distribution. The subscription is a pre-purchase mechanism, not a fundamentally different economic model.

CinePass Tier	Monthly Price & Allocation
CinePass Standard	\$12/month — 80 CineCredits (~6 standard rentals)

CinePass Plus	\$22/month — 180 CineCredits (~13 rentals + 1 ownership token)
CinePass Collector	\$45/month — 420 CineCredits + 1 free Collector Token per quarter

6.4 Smart Contract Revenue Distribution

Every content upload on CineChain requires the rights holder to define a revenue split contract at the time of registration. This contract specifies the wallet addresses of all revenue recipients and their respective percentage entitlements. The split can accommodate an unlimited number of recipients — production company, co-producers, talent residuals pool, sales agent, and protocol fee — and is immutable once registered unless a formal amendment is submitted and co-signed by all parties listed in the original contract.

Transaction Flow

1. Audience member redeems CineCredits to purchase an FLT.
2. CineCredits are converted to the platform's settlement token (USDC) at the point of redemption.
3. The settlement amount is routed to the film's revenue distribution smart contract.
4. The contract executes atomically, distributing USDC to all defined recipient wallets in the specified proportions.
5. The distribution event is emitted as a blockchain transaction, publicly verifiable by all parties.
6. All recipients can view their real-time balance through the CineChain dashboard or directly via blockchain explorer.

Example Revenue Split — Independent Feature Film

Production Company (primary rights holder): 65%
 Co-Producer A: 12%
 Co-Producer B: 8%
 Talent Residuals Pool (SAG-AFTRA equivalent): 10%
 CineChain Protocol Fee: 5%

A \$10 Rental Token purchase at 1 CineCredit = \$0.10 (100 credits) distributes:
 \$6.50 → Production Company | \$1.20 → Co-Producer A | \$0.80 → Co-Producer B
 \$1.00 → Talent Residuals | \$0.50 → Protocol — all in the same transaction block

6.5 Layered Watermarking and Piracy Attribution

CineChain's anti-piracy architecture operates at two distinct and complementary levels. Together they create a forensic chain of custody that is more robust than any single-layer DRM or watermarking approach currently available to independent distributors.

Layer 1 — Session-Level Perceptual Watermark

At the moment of decryption and delivery, a unique perceptual watermark is embedded into the video frames in real time. This watermark encodes the viewer's wallet address, device fingerprint, session timestamp, and token identifier into the video signal using a spread-spectrum technique that distributes the watermark information across thousands of frames. The watermark is imperceptible during normal viewing at all standard display sizes and frame rates.

Critically, the watermark is designed to survive the most common piracy extraction techniques: screen recording, re-encoding to different codecs, resolution reduction, compression, and social media upload transcoding. Watermark extraction algorithms can identify the embedded identifier even from a 360p social media re-upload of a 4K original. When a pirated copy is detected, the extraction process returns the specific session record that produced it — identifying the account, device, location, and viewing timestamp of the source within minutes.

Layer 2 — Blockchain Content Hash

The SHA-3-256 hash of the original master file, registered on-chain at the time of upload, provides a second layer of forensic infrastructure. If a watermark is detected in a pirated copy but the token holder disputes responsibility — claiming, for example, that the watermark was inserted post-hoc or that the copy originated from a different source — the content hash provides independent verification. Any copy that matches the on-chain hash, combined with a detectable session watermark, constitutes cryptographically verifiable evidence of the chain of custody from the authenticated viewer session to the pirated copy.

Enforcement Workflow

- CineChain's automated monitoring service (PiracyRadar module) continuously scans torrent indexers, Telegram channels, cyberlockers, and social platforms using perceptual hash matching.
- Upon detection, the system extracts the session watermark from the unauthorized copy.
- The session record is retrieved and cross-referenced with the viewer account.
- An automated DMCA takedown is dispatched to the hosting platform.
- The rights holder is notified with full forensic detail and optionally triggers account suspension per the platform's terms of service.
- For repeat violations or large-scale redistribution, the forensic package is compiled for legal action.

7. Governance — DAO Architecture

CineChain is governed as a decentralized autonomous organization (DAO) from the first day of mainnet launch. This is not a roadmap commitment to future decentralization — it is a day-one architectural requirement, because the platform's value proposition to filmmakers depends on a credible, structurally enforced guarantee that no single entity can capture or alter the protocol's rules to serve their interests at the expense of creators.

7.1 Governance Token — CINE

CINE is the protocol's governance token. It is not a speculative asset and is explicitly designed to minimize secondary market trading incentives. CINE is earned through verified contributions to the protocol ecosystem and is non-transferable for the first 24 months post-launch. Earning mechanisms include verified content uploads (rights holders), sustained viewing activity (audiences), protocol development contributions (engineers), and curatorial activity (curators). CINE cannot be purchased directly on the open market during the initial governance period.

7.2 Voting Mechanics

Governance proposals are submitted on-chain and enter a seven-day community review period before proceeding to a five-day voting window. Proposals require a minimum quorum of 15% of circulating CINE to be valid and a simple majority for standard protocol changes. Changes to core economic parameters — protocol fee rate, token tier definitions, and revenue distribution mechanics — require a supermajority of 67% and a 30-day review period. Emergency security proposals can be fast-tracked through a 48-hour process with a 75% approval threshold.

7.3 Scope of Governance

- Protocol fee rate adjustments
- Token tier definitions and pricing band updates
- New territory rights frameworks
- Curatorial organization accreditation
- Smart contract upgrade approvals
- Treasury allocation for grants and development bounties
- Content standards and prohibited content policies

8. Discovery & Curation Layer

CineChain's discovery architecture is built on the conviction that algorithmic recommendation systems, while powerful, are structurally misaligned with the needs of independent film because they optimize for engagement signals that favor already-popular content. The platform replaces centralized algorithmic curation with a verifiable, reputation-staked community curation system.

8.1 Curatorial Organizations

Any film festival, critical organization, film school, or subject-matter community can apply for accredited curator status on CineChain. Accredited curators can issue on-chain endorsement badges to films in the catalog. These badges are permanent, publicly verifiable, and tied to the curator's on-chain reputation score. Audiences follow curators whose taste they trust, and the curated collections of followed organizations appear prominently in their discovery feed.

8.2 Reputation Scoring

Curator reputation is scored based on the engagement and completion rates of films they endorse, the diversity and volume of their endorsements, and community ratings from viewers who discovered a film through their recommendation. High-reputation curators receive greater visibility in the discovery interface and are eligible for a small revenue share on tokens purchased through their curatorial referral — a direct economic alignment between curation quality and curation reward.

8.3 Festival Integration

Film festivals can mirror their official selection on CineChain as a digital companion to their physical event. A festival selection badge from a recognized institution (Sundance Official Selection, IDFA Competition, etc.) carries significant curatorial weight in the discovery layer and directly increases a film's visibility to the relevant audience segment. Festivals receive a configurable revenue share on digital sales made through their branded page on the platform.

9. Technology Stack

9.1 Blockchain Layer

Component	Technology & Rationale
Primary Blockchain	Ethereum Layer 2 — Polygon zkEVM. Provides Ethereum-level security with dramatically lower gas costs, enabling economical micro-transactions for credit redemptions.
Smart Contract Language	Solidity 0.8.x — industry standard, extensive security tooling (OpenZeppelin), large auditor ecosystem.
Token Standard	ERC-1155 for FLTs (supports fungible and non-fungible in a single contract). ERC-20 for CINE governance token.
Royalty Standard	EIP-2981 for on-chain royalty enforcement across compatible secondary marketplaces.
Content Registry	Custom Solidity registry contract mapping content hashes to rights holder addresses and metadata URIs.
DAO Framework	OpenZeppelin Governor — battle-tested, audited governance contract framework with configurable quorum and timelock.
Settlement Currency	USDC (Circle) — regulatory-compliant, USD-pegged stablecoin for all revenue distributions.
Indexing	The Graph Protocol — decentralized indexing of on-chain events for real-time dashboard analytics.

9.2 Backend Infrastructure

Component	Technology & Rationale
API Layer	Node.js with Fastify — high-throughput REST and WebSocket API, chosen for its low overhead and plugin ecosystem.
Content Storage	IPFS + Filecoin for decentralized redundant storage of encrypted film files. Pinata as the IPFS pinning service provider with geographic redundancy.
Key Management	HashiCorp Vault for Content Encryption Key (CEK) management. HSM-backed key storage for master keys.
Watermarking Engine	Custom C++ library compiled to WebAssembly for client-side watermark embedding at decode time. Server-side Python service for watermark extraction and forensic analysis.
Streaming Delivery	HLS (HTTP Live Streaming) with AES-128 segment encryption. CDN delivery via Cloudflare Stream for global low-latency delivery.

Database	PostgreSQL for transactional data (user accounts, credit balances, session records). Redis for session caching and rate limiting.
Search and Discovery	Elasticsearch for full-text and faceted search across the film catalog. Collaborative filtering model (PyTorch) for personalized recommendations within curated collections.
Monitoring	Grafana + Prometheus for infrastructure observability. Datadog APM for application performance monitoring.

9.3 Frontend Stack

Component	Technology & Rationale
Web Application	Next.js 14 (React) with TypeScript — server-side rendering for SEO and first-load performance, App Router for streaming UI patterns.
Mobile Applications	React Native — code-sharing across iOS and Android, access to native DRM APIs (Widevine L1, FairPlay) for secure streaming.
Wallet Integration	WalletConnect v2 + RainbowKit for optional crypto-native wallet connection. Privy.io for embedded wallets for users who prefer email/social login.
UI Component Library	Radix UI (headless) + Tailwind CSS — accessible, unstyled primitives with a custom CineChain design system applied.
Video Player	Video.js with HLS.js and custom DRM plugin for encrypted stream playback. WASM watermarking module loaded as a player plugin.
State Management	Zustand for client-side state. React Query (TanStack) for server state and caching.

9.4 DevOps and Infrastructure

Component	Technology & Rationale
Container Orchestration	Kubernetes (GKE) for production workload management with horizontal auto-scaling.
CI/CD	GitHub Actions for automated testing, smart contract compilation, and deployment pipelines.
Smart Contract Testing	Hardhat + Foundry for unit testing, integration testing, and gas optimization analysis.
Security Auditing	Pre-mainnet third-party audit (Trail of Bits or equivalent). Immunefi bug bounty program post-launch.
Infrastructure as Code	Terraform for reproducible cloud infrastructure provisioning across GCP and Cloudflare.

Secrets Management	Google Secret Manager for runtime secrets. Hardware Security Module (HSM) for master key material.
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10. System Architecture

10.1 High-Level Architecture

CineChain's architecture follows a three-tier model separating the presentation layer, the application services layer, and the data and blockchain layer. All tiers are independently scalable and communicate through well-defined API contracts.

Architecture Flow — Film Upload

Filmmaker → Upload Dashboard (Next.js) → Upload API (Fastify) → Content Validation Service → AES-256-GCM Encryption (KMS) → IPFS/Filecoin Storage → SHA-3-256 Hash Computation → Polygon zkEVM Registry Contract (on-chain registration) → Metadata Index (Elasticsearch) → Dashboard confirmation with on-chain transaction receipt

Architecture Flow — Token Purchase & Streaming

Viewer → Storefront (Next.js) → Payment API → CineCredit Deduction (PostgreSQL) → FLT Smart Contract (Polygon zkEVM) → Token Minted to Viewer Wallet → Revenue Distribution Contract (atomic USDC split to all recipients) → Session Token Issued → CDN Signed URL Generated → HLS Stream Initiated → WASM Watermark Module (browser/native) → Viewer-specific perceptual watermark embedded per segment → Session record logged (PostgreSQL + on-chain event)

10.2 Watermarking Architecture — Technical Detail

The watermarking pipeline is the most technically differentiated component of CineChain's architecture. It is designed to be imperceptible, resilient to common extraction attacks, and computationally lightweight enough to execute at decode time without introducing perceptible latency.

Embedding Phase

The watermarking algorithm operates in the frequency domain using a variant of spread-spectrum watermarking. For each authorized viewing session, a unique session identifier (a 128-bit value derived from the viewer's wallet address, device fingerprint, session timestamp, and a random nonce) is encoded into a pseudo-noise sequence using a secret key known only to the platform. This sequence is then modulated into the discrete cosine transform (DCT) coefficients of selected frames throughout the video — distributed across the film's entire runtime rather than concentrated in any single section, making partial removal impractical without visible degradation.

The embedding is performed by a WebAssembly module loaded within the video player, which receives the session identifier as an encrypted payload from the server at the start of each authorized session. The WASM module decrypts the identifier client-side, generates the watermarking sequence, and applies it to decoded video frames before rendering to the display buffer. The watermark is never visible in network traffic — the encrypted HLS stream carries no watermark; it is applied only in the final decode stage within the trusted execution environment of the player.

Extraction and Detection Phase

The extraction service maintains the same secret key used in embedding. Given a sample of video from a suspected unauthorized copy, the extraction algorithm applies a correlation test between the received signal and the expected pseudo-noise sequences for each possible session identifier. A positive correlation above a defined statistical threshold identifies the session. The algorithm is optimized to return a confident identification from as few as 30 seconds of video content at 360p resolution, making it effective against heavily compressed social media re-uploads.

10.3 Key Management Architecture

Content Encryption Keys (CEKs) are the most sensitive material in the CineChain key hierarchy. Each film has a unique CEK generated at upload time. The CEK is wrapped using the platform's master key, which is stored in a Hardware Security Module (HSM) within HashiCorp Vault's managed service. At the point of authorized viewing, the CEK is unwrapped and delivered to the streaming session over a mutually authenticated TLS channel — it exists in plaintext only within the platform's secure computing environment and is never stored or logged in cleartext. Session tokens authorize a single viewing session and expire after the token's defined access window.

11. Data Models

11.1 Core On-Chain Data Structures

ContentRegistry Smart Contract — Key Fields

contentId: bytes32 (SHA-3-256 hash of unencrypted master file)
rightsHolder: address (primary wallet)
metadataURI: string (IPFS URI of JSON metadata package)
registrationTimestamp: uint256
territoryBitmap: uint256 (bitmask of authorized territories)
revenueContractAddress: address (linked split contract)
isActive: bool

RevenueDistribution Smart Contract — Key Fields

contentId: bytes32 (foreign key to ContentRegistry)
recipients: address[] (ordered list of recipient wallets)
shares: uint256[] (basis points per recipient, must sum to 10000)
resaleRoyaltyBps: uint256 (Collector Token resale royalty, typically 1000 = 10%)
totalDistributed: uint256 (cumulative USDC distributed, all time)
lastDistributionTimestamp: uint256

FilmLicenseToken (ERC-1155) — Token ID Structure

Token ID encoding: [contentId (128 bits)] + [tier (8 bits)] + [edition (56 bits)] + [sequence (64 bits)]
Tier 0: Rental Token — expires 48 hours after first play
Tier 1: Ownership Token — perpetual, transferable once
Tier 2: Collector Token — perpetual, fully transferable, EIP-2981 royalties

11.2 Off-Chain Data Models (PostgreSQL)

Table	Key Fields
users	id, wallet_address, email (optional), credit_balance, kyc_status, created_at
films	id, content_id (on-chain ref), title, description, runtime, genres, languages, territories, upload_status, ipfs_cid, created_at

sessions	id, user_id, film_id, token_id, device_fingerprint, ip_address, session_key_hash, started_at, watermark_id, status
transactions	id, user_id, film_id, token_tier, credits_spent, tx_hash (on-chain), created_at
watermarks	id, session_id, wallet_address, device_fingerprint, timestamp, nonce, status
curator_endorsements	id, curator_id, film_id, badge_type, on_chain_tx_hash, issued_at
piracy_detections	id, film_id, detected_at, source_url, extracted_session_id, watermark_confidence, status

12. Security Architecture

12.1 Threat Model

CineChain's threat model identifies four primary attack surfaces: smart contract vulnerabilities, key management compromise, watermark circumvention, and platform-level fraud. Each is addressed through distinct countermeasures.

Threat	Countermeasure
Smart contract reentrancy / overflow	OpenZeppelin battle-tested libraries, Checks-Effects-Interactions pattern, pre-launch third-party audit (Trail of Bits), Immunefi bug bounty
CEK exfiltration from streaming session	WASM sandbox execution, session-scoped key derivation, no CEK persistence on client, certificate pinning in mobile apps
Watermark stripping / circumvention	Spread-spectrum embedding (resilient to re-encoding), multiple independent watermark layers, detection threshold calibrated for partial destruction
Credential stuffing / account takeover	Rate limiting on auth endpoints, hardware-based 2FA for filmmaker accounts, anomalous session detection, wallet-based authentication as primary method
Sybil attacks on governance	Non-transferable CINE during lock period, proof-of-contribution earning model, one-human-one-vote identity layer (optional Worldcoin integration for high-stakes votes)
USDC smart contract failure	Multi-signature emergency pause capability, 48-hour settlement delay for amounts above \$10,000, protocol reserve fund covering 90 days of distributions

12.2 Audit and Compliance

All smart contracts undergo a formal third-party security audit prior to mainnet deployment. The audit scope includes the ContentRegistry, RevenueDistribution, FilmLicenseToken, and DAO Governor contracts. Audit reports are published publicly on the CineChain GitHub repository. A continuous bug bounty program on Immunefi offers rewards up to \$500,000 for critical vulnerabilities in deployed contracts. The platform complies with GDPR for European users (right to erasure applies to off-chain data; on-chain records are pseudonymous by design), COPPA for minor protection, and applicable territorial content regulations through the territory bitmap access control system.

13. API Design

13.1 REST API — Core Endpoints

Endpoint	Description
POST /api/v1/films/upload	Initiates multipart upload for film file and metadata. Returns upload session ID and pre-signed IPFS upload URLs.
POST /api/v1/films/{id}/register	Triggers on-chain registration after upload completes. Returns transaction hash.
GET /api/v1/films/{id}	Returns full film metadata, available token tiers, pricing, and curatorial endorsements.
POST /api/v1/tokens/purchase	Redeems CineCredits for an FLT. Returns token ID, streaming session URL, and tx hash.
GET /api/v1/stream/{sessionId}	Returns signed HLS manifest URL for authorized session. Valid for session duration only.
GET /api/v1/dashboard/revenue	Returns real-time revenue breakdown for authenticated rights holder.
POST /api/v1/credits/purchase	Initiates credit purchase via payment rail. Returns credit balance update.
GET /api/v1/discover	Returns personalized discovery feed based on followed curators and viewing history.
POST /api/v1/governance/propose	Submits a governance proposal on-chain.
GET /api/v1/piracy/detections	Returns piracy detection events for authenticated rights holder.

13.2 WebSocket Events

Event	Description
revenue:distribution	Real-time notification to rights holder when a distribution event fires on-chain.
piracy:detected	Real-time alert to rights holder when a piracy detection is confirmed.
governance:vote	Live vote count updates during active governance proposals.
stream:heartbeat	Session keep-alive and watermark acknowledgment between player and server.

14. Product Roadmap

Phase 1 — Hackathon MVP (Day 1–2)

The Hackathon MVP demonstrates the complete end-to-end transaction in a single coherent live flow. This is the proof-of-concept that validates every core claim of the platform in a demonstrable environment.

- Filmmaker uploads a short film via a simplified dashboard
- Film is AES-256 encrypted and registered on Polygon Mumbai testnet with SHA-3-256 hash
- Test user browses catalog, purchases a Rental Token using mock CineCredits via UI
- FLT smart contract mints token and executes revenue split to 3 pre-defined wallet addresses
- Film streams via HLS with WASM session watermark applied in real time
- Watermark extraction demonstration on a captured frame confirms attribution
- Revenue dashboard shows real-time distribution event

Phase 2 — Alpha Launch (Months 1–4)

- Mainnet deployment on Polygon zkEVM
- CineCredit purchase via Stripe (credit card)
- Full watermarking pipeline with PiracyRadar automated scanning
- Filmmaker dashboard with real-time revenue analytics
- 3–5 festival partnerships for curated launch catalog (target: 50–100 short films and documentaries)
- Web application only (no mobile)
- DAO governance deployed with founding contributor CINE allocation

Phase 3 — Beta Expansion (Months 5–9)

- iOS and Android mobile applications with native DRM (Widevine L1, FairPlay)
- CinePass subscription tier
- Collector Token secondary market
- Curator badge system with 10+ accredited curatorial organizations
- Feature film support (DCP ingest, 4K HDR delivery)
- Institutional licensing for film schools and archives
- International payment localization (INR, EUR, BRL, NGN)

Phase 4 — Protocol Maturity (Months 10–18)

- Open API for third-party storefronts built on CineChain protocol
- Filmmaker SDK for embedding CineChain purchase flows in personal websites
- Advanced analytics: audience demographic insights for rights holders (privacy-preserving, aggregated)
- Expansion to live event streaming with per-session tokenization
- Protocol v2 governance proposal: cross-chain token portability (Ethereum mainnet, Solana)
- Grant program for emerging filmmakers funded by DAO treasury

15. Success Metrics & KPIs

Metric	6-Month Target	18-Month Target
Films on platform	500	10,000
Registered filmmakers / distributors	200	3,000
Registered audience members	5,000	250,000
Monthly token transactions	1,000	100,000
Monthly revenue distributed (USD)	\$20,000	\$2,000,000
Average protocol fee collected monthly	\$1,000	\$100,000
Piracy detections actioned	N/A	500+
Accredited curatorial organizations	5	50
DAO governance participation rate	25%	40%
CinePass subscribers	100	15,000

16. Risks & Mitigations

Risk	Mitigation Strategy
Smart contract exploit causing loss of distributed funds	Pre-launch audit, formal verification of revenue distribution contract, 48-hour settlement delay on large amounts, protocol insurance fund
Regulatory classification of FLTs as securities	Legal opinion obtained pre-launch. Rental and Ownership Tokens structured as consumption goods, not investment instruments. Collector Tokens carry full KYC/AML compliance. Securities counsel engaged in US, EU, and Singapore jurisdictions.
Watermark circumvention by sophisticated actors	Multi-layer architecture (session + content hash). Watermark algorithm regularly updated via protocol upgrades. Secondary detection via behavioral analytics (anomalous session patterns).
Cold start — insufficient content at launch	Festival partnership strategy ensures 50–100 curated titles at launch. Filmmaker grant program incentivizes early uploads. Exclusive premiere window negotiated with 3 festival partners.
UX complexity deterring mainstream adoption	CineCredit layer fully abstracts blockchain. Privy.io embedded wallets eliminate wallet setup for email/social login users. Usability testing with non-technical audience panels in Alpha phase.
Competitor response from major platforms	Open-source protocol cannot be acquired or shut down. Network effects accumulate in the protocol (on-chain content registry, established curator reputation graph). Platform is the reference implementation, not the moat.
IPFS/Filecoin availability for film delivery	CDN caching layer (Cloudflare) serves 99%+ of requests. IPFS/Filecoin is source of record, not primary delivery path. Geographic redundancy across 3+ Filecoin storage providers.
DAO governance capture by large token holder	CINE non-transferable lock period prevents accumulation. Quadratic voting considered for Phase 3 governance upgrade. Founding team CINE vests over 4 years with 1-year cliff.

17. Competitive Analysis

CineChain occupies a unique position in the distribution landscape because no existing platform combines all four of its core properties in a single product. The relevant competitive landscape spans four categories.

Platform	Overlap with CineChain	Key Differentiation
Vimeo On Demand	Per-title purchase and rental for independent film	No blockchain, no watermarking, no smart contract revenue, opaque analytics, 10% fee
MUBI	Curated independent and arthouse film	Subscription only, no ownership model, centralized curation, no creator revenue transparency
Audius	Decentralized creator distribution (audio)	Music only, no video infrastructure, no DRM/watermarking, limited curatorial structure
Veritone / Verimatrix	Enterprise watermarking and content protection	B2B enterprise, not a distribution platform, no blockchain revenue model, prohibitively expensive for indie
Mirror.xyz	Creator-owned publishing with token-gated content	Text/writing focused, no video delivery infrastructure, no film industry integrations
OpenSea / NFT platforms	Secondary market for digital collectibles	No content delivery, no DRM, no streaming, speculation-focused, no filmmaker tooling

CineChain's defensible advantage is the combination of three properties that no competitor currently holds simultaneously: forensic attribution (making piracy traceable), transparent on-chain revenue distribution (making intermediary opacity structurally impossible), and open-source governance (making platform capture structurally impossible). Any one of these properties alone is replicable. The combination, built into a single coherent protocol, creates a compounding network effect that strengthens with each additional film registered and each additional curator credentialed.

18. Business Model

18.1 Revenue Streams

Revenue Stream	Description & Rate
Protocol fee on token purchases	5% of all FLT purchase transactions. Applied at smart contract level, non-negotiable. Collected in USDC and routed to DAO treasury.
Protocol fee on secondary sales	2.5% of all Collector Token secondary market transactions, in addition to the rights holder's resale royalty.
CinePass subscription revenue	A portion of monthly CinePass subscription fees is retained as platform revenue after credit allocation to subscribers.
Institutional licensing revenue	Film schools and archives pay a platform access fee for bulk licensing infrastructure, over and above per-title smart contract distributions.
Developer API tier	Third-party developers building storefronts on the protocol pay a tiered API access fee based on transaction volume.

18.2 Treasury and Protocol Sustainability

All protocol fees are collected in the DAO treasury, denominated in USDC. Treasury funds are allocated through quarterly governance votes across three categories: engineering and protocol development (target 50%), filmmaker grants and ecosystem development (target 30%), and protocol reserve fund for security and operational continuity (target 20%). The protocol is designed to be self-sustaining from protocol fees alone at a transaction volume of approximately 50,000 monthly token purchases — a target achievable within 18 months based on comparable platform growth benchmarks.

19. Open Source Strategy

CineChain's open-source architecture is its most strategically significant long-term differentiator. The complete smart contract codebase, the watermarking extraction library (the embedding algorithm remains proprietary to prevent circumvention), the filmmaker SDK, and the API specification are published under a permissive open-source license (MIT or Apache 2.0) on day one.

This serves three distinct strategic purposes. First, it creates an auditable trust foundation — any filmmaker, technical advisor, or investor can independently verify that the revenue distribution contracts execute exactly as documented. Second, it enables a developer ecosystem — studios, festivals, film schools, and third-party developers can build their own storefronts, tools, and integrations on the protocol without permission or commercial negotiation. Third, it makes the platform structurally unkillable — even if the founding team dissolves, the protocol persists, and any capable team can operate or fork it.

The reference implementation of the CineChain platform (the web application, mobile apps, and filmmaker dashboard) is maintained by the founding team and funded through the DAO treasury. It competes in the open market with any third-party implementations built on the protocol — and is expected to win on the basis of product quality and developer resources, not exclusivity.

20. Appendices

Appendix A — Glossary

Term	Definition
FLT (Film License Token)	An ERC-1155 blockchain token representing a specific viewing right for a registered film on CineChain.
CEK (Content Encryption Key)	The unique AES-256 symmetric key used to encrypt a specific film's content.
Perceptual Watermark	An imperceptible modification to video frame data that encodes a unique session identifier, resilient to re-encoding.
CineCredit	CineChain's platform-level stable unit of account, pegged to USD at \$0.10 per credit.
CinePass	CineChain's monthly subscription tier providing pre-allocated credit bundles.
DAO	Decentralized Autonomous Organization — a blockchain-governed community with on-chain voting rights.
CINE	CineChain's governance token, earned through protocol contribution and used for DAO voting.
DCP (Digital Cinema Package)	The standard file format for theatrical film distribution to digital cinema projectors.
ERC-1155	Ethereum token standard supporting both fungible and non-fungible tokens in a single contract.
EIP-2981	Ethereum standard for on-chain royalty enforcement on secondary token sales.
USDC	USD Coin — a regulated, USD-pegged stablecoin issued by Circle, used for all revenue distributions on CineChain.

Appendix B — References

- Polygon zkEVM Documentation: <https://zkevm.polygon.technology>
- OpenZeppelin Governor Framework: <https://docs.openzeppelin.com/contracts/governance>
- EIP-2981 Royalty Standard: <https://eips.ethereum.org/EIPS/eip-2981>
- ERC-1155 Multi-Token Standard: <https://eips.ethereum.org/EIPS/eip-1155>
- HLS Specification (RFC 8216): <https://datatracker.ietf.org/doc/html/rfc8216>
- Filecoin Documentation: <https://docs.filecoin.io>
- HashiCorp Vault Key Management: <https://www.vaultproject.io>
- The Graph Protocol: <https://thegraph.com/docs>

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